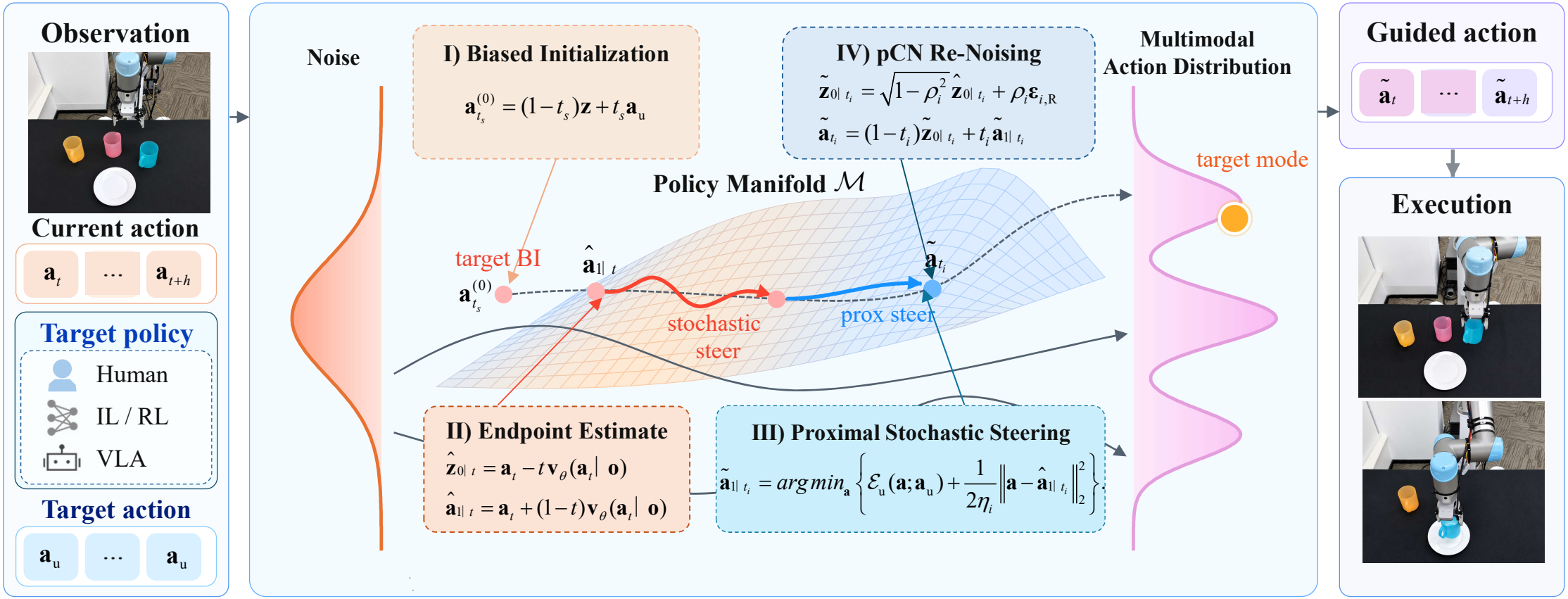

Observation
Current action
$\mathbf{a}_t$ ⋯ $\mathbf{a}_{t+h}$
Target policy
Human
IL / RL
VLA
Target action
$\mathbf{a}_u$ ⋯ $\mathbf{a}_u$
Noise
I) Biased Initialization
$\mathbf{a}_{t_s}^{(0)} = (1-t_s)\mathbf{z} + t_s\mathbf{a}_u$
IV) pCN Re-Noising
$\tilde{\mathbf{z}}_{0|t_i} = \sqrt{1-\rho_i^2}\,\hat{\mathbf{z}}_{0|t_i} + \rho_i \boldsymbol{\varepsilon}_{i,R}$
$\tilde{\mathbf{a}}_{t_i} = (1-t_i)\tilde{\mathbf{z}}_{0|t_i} + t_i\tilde{\mathbf{a}}_{1|t_i}$
Multimodal Action Distribution
target mode
Policy Manifold $\mathcal{M}$
target BI
$\mathbf{a}_{t_s}^{(0)}$
$\hat{\mathbf{a}}_{1|t}$
stochastic steer
prox steer
$\tilde{\mathbf{a}}_{t_i}$
II) Endpoint Estimate
$\hat{\mathbf{z}}_{0|t} = \mathbf{a}_t - t\mathbf{v}_\theta(\mathbf{a}_t|\ \mathbf{o})$
$\hat{\mathbf{a}}_{1|t} = \mathbf{a}_t + (1-t)\mathbf{v}_\theta(\mathbf{a}_t|\ \mathbf{o})$
III) Proximal Stochastic Steering
$\tilde{\mathbf{a}}_{1|t_i} = \arg\min_{\mathbf{a}} \left\{ \mathcal{E}_u(\mathbf{a};\mathbf{a}_u) + \frac{1}{2\eta_i} \left\| \mathbf{a} - \hat{\mathbf{a}}_{1|t_i} \right\|_2^2 \right\}$
Guided action
$\tilde{\mathbf{a}}_t$ ⋯ $\tilde{\mathbf{a}}_{t+h}$
Execution